FinanceAsia

Ready money

The internet has been crying out for its own currency since the 1990s and Bitcoin offers a solution that could be quick, cheap and verifiable.

China's announcement in December that it was banning local financial institutions from using bitcoins caused a collapse in the price of the digital currency and led a chorus of talking heads to predict its demise.

Things don't seem to have improved much since then. Japan's Mt Gox, the biggest bitcoin exchange, went bust in February after the unexplained loss of 850,000 bitcoins (valued at the time at more than $450 million) to hackers. Then in late March the US said that it would tax Bitcoins as property.

But these pale compared to the China ban. After Beijing's statement, speculators watched the value of their digital portfolios fall in half during the last few weeks of 2013 and, despite a brief rally in January, the bitcoin slump has continued (prices diverge across multiple different exchanges, but the peak value was roughly $1,100 compared to less than $500 today).

The joint statement from China's banking, securities and insurance regulators was unequivocal: "Although [bitcoin] is called currency, it is not issued by a monetary authority, it does not have the status of legal tender and obliged payment status of currency, it is not currency in the true sense. Bitcoin is a specified virtual commodity, it does not have equal legal status with currency, and it cannot and should not be circulated as currency on the market."

It went on to say, in effect, that financial institutions must not offer any products or services related to bitcoin, casting doubt on the future of virtual currencies in its biggest market.

So did China kill bitcoin? Not according to Dave Chapman, co-founder and chief operating officer of ANX, a Hong Kong-based exchange that offers trading in bitcoin and other digital currencies — and which unveiled the city's first bitcoin ATM in February.

"Whilst the media adequately relayed this message [that PBoC had barred financial institutions from transacting in bitcoins], the majority of the articles published failed to mention that in the same ruling it was confirmed that private individuals are allowed to trade bitcoin at their own risk," he said during an interview at the exchange. "That's a win for us."

Indeed, the statement was like a warm embrace to bitcoin professionals in Asia. Far from "banning" bitcoin, the regulators acknowledged it for the first time and started to lay the foundations for better regulation. They even said that financial and payment organisations should inform the public about virtual goods and currencies. This is by no means a ban.

Other stories about countries banning bitcoin betray a similar reality: regulators across the region are accepting that bitcoin is significant enough to warrant their attention — even if they do not always seem to have much to say.

End of the beginning

This regulatory acknowledgement could be the first of many big steps that bitcoin takes in 2014. As the product matures, a new breed of more professional operations is slowly edging out cowboy operators.

In response to Mt Gox's "tragic violation of trust", for example, a group of top bitcoin professionals put out a joint statement condemning its shoddy approach and providing a united and credible voice for the industry.

"Strong bitcoin companies, led by highly competent teams and backed by credible investors, will continue to thrive and to fulfil the promise that bitcoin offers as the future of payment in the internet age," said the statement, signed by the co-founder of Coinbase and the chief executives of Kraken, Bitstamp.net, BTC China, Blockchain.info and Circle.

If anything, players such as Mt Gox were becoming a pain for those who wanted bitcoin to break through to mainstream acceptance. Although it grew to become the world's most popular bitcoin exchange, the code behind Mt Gox had been written in just a few days and the exchange operated without any of the typical checks involved in the regulated financial industry.

"Mt Gox was one of the reasons we started," said Chapman. "We thought we could do a better job."

Firms like ANX are now leading the new generation of bitcoin exchanges — its three founders are all serious professionals. Chapman joined in November from HSBC, where he had been responsible for the derivatives development team in Hong Kong after a career that included stints at Bear Stearns, Credit Suisse, Barclays and ABN Amro.

Hugh Madden was a fellow HSBC banker, while the chief executive

Ken Lo worked in the telecoms industry.

Registered members must go through proper identity checks and the exchange takes anti-money laundering and know-your-customer provisions seriously. Joining the exchange is still an entirely online process, though customers have been known to show up at the Wan Chai office with wads of cash.

ANX's customer funds are all held in "cold storage", which means they are offline and secure from hackers.

This process of weeding out the weak links is an important one, according to Dmitry Dain, chief technology officer and founder at Betalabs, who was interviewed as part of a Goldman Sachs research report.

"As legitimacy of bitcoin grows, the assumption is that more investment will flow into the operation of the exchanges and that they will become more stable," he said. "But it is still a Wild West out there for the exchanges."

Attitudes can change quickly. Just six months ago most people reckoned that the closure of Silk Road, an online black market, would lead to a dramatic drop in bitcoin usage as criminals reverted to dark alleys and bank-issued currency. The opposite was true: bitcoin VALUES IN DOLLAR TERMS shot up by more than 400% immediately after the FBI closed the website in October.

The more professional exchanges also noticed a flight to quality after Mt Gox went down. "We had our busiest week ever," said Chapman.

US retailers such as Zynga, Overstock and TigerDirect have all recently started to accept bitcoins and more are joining them. CoinMap, a website that claims to map retail outlets around the world that accept bitcoin and litecoin payments, lists more than 150 stores in Asia — and close to 4,000 worldwide. "That would mean bitcoin prices going through the roof," he predicted.

The next step, which Chapman is confident will happen this year, is for a major online player such as Amazon or Google to start accepting them.

"This would naturally result in much wider appreciation of bitcoin as a payment technology — and it would most certainly increase the demand for bitcoin," he predicted.

Early adopters can already use bitcoin to pay for things on Amazon (as well as Walmart, Macy's and Target) by installing a simple Google Chrome web plugin that uses the retailer's API, an interface that allows app developers to offer third-party services. It also searches the internet for cheaper deals and seamlessly finds discounts. Such tools come with a cost because they involve buying through a third-party reseller, which can void warranties on some products. Going direct through Amazon, rather than a less trusted app developer, would still offer a better experience for most customers.

Whether that will ever happen is the big question. Google appeared to say in February that it was exploring options to integrate Bitcoin after an email from the commerce team was circulated in public. The company later dialled back the statement, but netizens are now convinced that Google is working on it. That seems a reasonable assumption.

Google's biggest revenue source is from online advertising. Using Bitcoin in this environment would mean AdSense payments could go through instantly instead of being sent by secure cheque (often internationally). A Gmail integration could allow anyone to send money securely by email, while a Bitcoin-enabled Google Wallet could revolutionise mobile payments.

No other large internet company has made a definitive statement about Bitcoin, but it is safe to assume that such innovative businesses are seriously considering the benefits of being a first-mover in this market.

There are challenges, of course. Google already spends a huge amount of time and effort keeping spammers and scammers out of its advertising business, but an anonymous payment system would hardly make that any easier. It would also presumably need to buy an exchange to be able to offer such a service.

Payday

None of the above helps to answer a simple bitcoin question that many people have: What's the point?

You can't pay your taxes with bitcoin, but its real selling point is not as a currency. In essence, it is a payment technology that allows for the cost-free exchange of assets between two anonymous parties without reliance on any central organisation, and which can be verified by anyone and cannot be reversed.

That probably doesn't sound very impressive to millennials out to change the world, but the truth is that existing payment technologies are much more expensive, much slower and much less secure. And bitcoin could be used to transfer any documents or assets, from securities to letters of credit.

Such a system would have broad applicability. It would offer small retailers huge advantages over credit cards, which cost them 2% to 4% of the sale and are commonly used fraudulently, resulting in chargebacks that leave the retailer out of pocket.

Overseas workers from the Philippines and other Asian countries who rely on money services such as Western Union would also benefit from cheaper fees. And their unbanked friends and relatives at home would only need a smartphone to create a mobile bank account.

For now, the conventional banking industry is sceptical. “If somebody can do all this risk-free, that would obviously be a positive for customers,” said Sridhar Kanthadai, head of transaction banking for North Asia at Standard Chartered. “The banking environment has regulatory challenges that are growing really fast, so it’s not easy to do at zero cost. Many of these ideas are very interesting, but regulators are getting nervous about it.”

But most of the banks that FinanceAsia contacted for this story, including several of the world’s top commercial banks, claimed not to be in a position to comment. Indeed, one of the world’s biggest and most sophisticated foreign exchange brokers told us: “It’s not a topic for us to discuss. I have double-checked our research database to see if there might be something helpful there to share with you but I'm afraid not at this stage.”

Even Goldman, which is the only bank we know to have published research on the subject, had to rely on external specialists for much of the content in its report.

That is not to say the banks have a duty to take virtual currencies seriously. It may of course turn out that Bitcoin is not a workable payments solution for wholesale banking clients. Doubtless the banks would prefer to “innovate” their own solutions that leverage the advantages of Bitcoin — transfers that are quick, cheap, verifiable and irreversible — at the same time as allowing some kind of fee structure.

Having said that, the zero-cost structure offers an important advantage that could be used for a variety of purposes. Charging a millionth of a bitcoin to receive emails, for example, could force spammers out of business. Such micropayments could also change the way that businesses monetise web traffic.

China's bitcoin trade volume at record levels

NICK FERGUSON, Contributing writer

HONG KONG -- Bitcoin trading in the Chinese currency is at record levels. New financial products and services, which offer zero-commission trading, margin loans and highly leveraged futures contracts, have ignited interest in the cryptocurrency.

A year ago, the price per bitcoin was in free fall and the People's Bank of China warned financial institutions and payment companies against using the cryptocurrency. The future looked murky.

Traders in March, however, bought and sold roughly 700 million yuan ($112 million) worth of bitcoin a day, according to website Bitcoinity. The record-level trading is more than double the volumes logged around a year ago.

China's central bank in late 2013 said financial and payment institutions were not allowed to conduct any sort of business using bitcoin. The bank's statement also said bitcoin did not have the same legal status as a currency. The central bank's moves essentially separated bitcoin from the mainstream financial industry. The PBOC, however, did not ban individuals from using and trading in bitcoin.

Getting creative

"Being creative Chinese entrepreneurs, we found workarounds," said Bobby Lee, CEO of Shanghai-based BTC China, now the world's biggest bitcoin exchange. BTC China and other exchanges now operate outside of mainstream finance. These dedicated bitcoin exchanges have seen trading bounce back.

Bitcoin is a cryptocurrency -- digital and decentralized, it relies on distributed computing power to clear and authenticate transactions. These transactions can be completed cheaply, immediately and without the need for escrow from a trusted third party such as a government, legal office or banking federation.

So far, however, payment transactions have represented a tiny part of bitcoin trading volumes. Activity is dominated by speculators. "We've had tractions and are getting more merchants to sign up, but even if payments grow 10,000% it will still be a drop in the bucket compared to exchange-traded products," said Lee.

Since 2014, local and international bitcoin exchanges have been luring Chinese investors with new ways for trading the cryptocurrency.

"Bitcoin came along and it just seemed like something completely new and fresh," said Matthew Ventura, a bitcoin speculator who operates five bitcoin ATMs in Macau. "I decided to give it a try and it's been very fun. It's exciting."

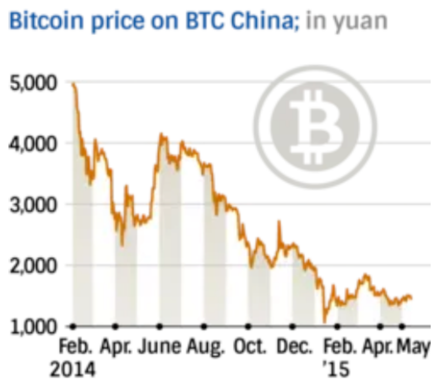
A gradual erosion of trading fees started in 2013 when BTC China scrapped its 0.3% commission. At the time, most exchanges were charging anywhere from 0.6% to 0.2% to swap bitcoin and yuan. Other exchanges followed BTC China's lead.

OKCoin, one of the most popular futures platforms, offers margin loans of up to twice a trader's account size and futures contracts with leverage of 10 or 20 times for a fee of 0.03%. Others offer near-identical terms.

All this has worked. China's interest in bitcoin has surged. That is despite the digital currency sinking from a peak of more than $1,100 in November 2013 to $230 today. Yuan values for bitcoin roughly correspond to those for dollars.

Good times

Worries about the future of bitcoin have subsided. "Trading volumes have risen over the past year, mostly due to the confidence that bitcoin is here to stay," said Lee.



"Bitcoin is not a hot topic within the Chinese government right now," said Roland Sun, a partner at Shanghai law firm Broad & Bright who specializes in cryptocurrencies. "They acted in 2013 because there was a Bitcoin's tiny market appears safe from Chinese regulators for now.

China's bitcoin trade volume at record levels - Nikkei Asia lot of volatility and it was high profile, but there are no indications that the government is going to take any further action on bitcoin. Government officials aren't very familiar with cryptocurrencies and would rather wait to see what the U.S. does."

The New York Department of Financial Services last July proposed a framework known as BitLicense that focuses on consumer protection, money laundering and cybersecurity rules for cryptocurrency businesses. It could form the basis of a bitcoin regulatory framework in the future.

Wall Street appears to think so. Goldman Sachs has published two lengthy reports on bitcoin, focusing on its usefulness as a payment protocol. The investment bank and IDG Capital Partners also in April led a $50 million investment in a U.S. bitcoin startup, Circle Internet Financial.

That, however, does not mean cryptocurrencies are about to go mainstream. The bitcoin market is tiny in comparison to conventional asset classes. The net buying quota available through Stock Connect, a trading link between the Shanghai and Hong Kong stock exchanges, is 10.5 billion yuan a day, much higher than the amount traded in bitcoin.

The small size of the bitcoin market and the potential of cryptocurrencies as legitimate financial tools appear to have pacified Chinese regulators for now.

Insurers wary as Singapore targets digital assets

Singapore is making a bid to become a regional hub for digital assets as China and India crack down on cryptocurrencies, but insurers remain wary of this new and volatile class of risks.

Earlier this week, the Monetary of Singapore licensed the city-state's first cryptocurrency exchange, which is operated by Australia's Independent Reserve. Dozens of others are reportedly waiting for approvals as interest in digital assets has increased this year amid soaring prices for cryptocurrencies and non-fungible tokens (NFTs).

The city was also at the centre of the biggest digital asset story of the year, when local entrepreneur Vignesh Sundaresan paid a record US$69 million (actually 42,329 in cryptocurrency ether) for an NFT by digital artist Beeple at a landmark Christie's auction.

Collectors who spend that kind of money on art would normally want to insure it (as we wrote about recently), but established insurance solutions for NFTs are not widely available.

"We have seen a few exploratory enquiries for NFTs in the past few months but appetite amongst specie underwriters is limited with only a handful in Lloyd's of London being able to offer some cover," Rhiannon Alban-Davies, Asia regional head of fine art and specie at Willis Tower Watson told InsuranceAsia News (IAN).

"We know that a few underwriters in Asia are exploring this and expect to see more solutions being offered albeit at low limits," she added. "Understanding the risk exposure and the basis of valuation, considering there have been few NFT sales on the secondary market, has been a challenge."

There is a market for digital artworks that are not on a blockchain platform and NFTs could ultimately end up being relatively easy to insure as they are extremely secure. Establishing valuations is the bigger challenge.

Cryptocurrencies

Valuations are also a challenge in the crypto sector. Prices of bitcoin and ether rose by hundreds of percent at the start of the year, only to collapse again after China cracked down on the crypto sector and indicated that it may be headed to a total ban, despite the country's prolific role in bitcoin mining. India is also considering legislation that will make it illegal to own private cryptocurrencies.

Such wild volatility makes this a difficult class of business to insure, but there is some capacity available for well-regulated crypto exchanges in the region — at a price.

In Australia, Independent Reserve charges A$35,000 (US$26,000) a year for accounts that offer insurance protection up to A$5 million. Now that it is licensed, it may start offering insurance in Singapore.

Munich Re is one (re)insurer that has been willing to underwrite such risks. Curv, a digital asset security platform bought by PayPal in May, announced a deal with the German company in 2019 to insure up to US$50 million in crypto assets stored in the cloud.

However, a spokesperson for Munich Re told IAN that its appetite for this kind of risk is very limited and that it offers small coverage elements only. "In terms of underwriting this risk, we take care to understand the business model behind the technical solution and exclude certain covers (eg, fraud, loss of digital assets)."

To highlight the challenge that volatility poses, a US$50 million portfolio of ether would have increased in value to around US$170 million today (or more than US$260 million in April).

Digital assets may eventually emerge as a significant class of business for insurers, but the underlying technology is already helping to modernise the industry.

Distributed ledger and blockchain technology can eliminate many of the legacy manual processes that make insurance slow, complicated and expensive.

Bharti AXA in India says that a motor insurance solution from Ledgertech, which is built on the Corda blockchain, has cut its costs per claim by 20%.

As the industry becomes more familiar with such solutions it may also become more comfortable with digital assets that leverage the same technology — assuming that valuations stabilise.

INSURANCE ASIA NEWS

Hyperconnecting reinsurance

Supercede co-founders Ben Rose and Jerad Leigh talk about giving the industry superpowers.

With a second lockdown renewal successfully completed in Japan on April 1, reinsurers and brokers have been patting themselves on the back for a job well done. And deservedly so.

But the industry is still a long way from being truly digital. Preparing, validating and compiling all the data to complete a reinsurance deal remains a time-consuming and tedious process, while sourcing risk or capacity is more challenging than ever.

"The way that deals are done hasn't fundamentally changed," says Ben Rose, co-founder and president of Supercede, a dedicated reinsurance technology company. "Everyone is very pleased with themselves because trading is happening, but it's still basically all being done through emails, zip files and spreadsheets. Replacing meetings with video calls does not make it much more efficient."

Rose and his co-founder Jerad Leigh spoke to InsuranceAsia News about their mission to modernise trading in the reinsurance industry through the creation of a "hyperconnected" reinsurance trading platform that truly harnesses technology to match risk with capital.

Supercede combines an established e-placement platform and reinsurance network with tools for data preparation. The goal is to make life easier for all existing market participants — cedants, brokers and reinsurers — rather than to disintermediate the broking function.

"We're broker-centric," says Leigh. "We've spent a lot of time talking to cedants and they actually value their brokers and the role that they play."

However, nobody values digging through vast spreadsheets in search of data for rekeying elsewhere. Streamlining these kinds of chores can dramatically reduce the amount of time it takes to close deals and empowers everyone involved to have much more productive interactions.

"Too often, brokers spend more time compiling data instead of doing deals," says Leigh, who is also the company's chief executive and was previously part of Aon Benfield's Japan team. "For example, the lead time for the April 1 renewal in Japan can start in September."

Leveraging technology to help cedants and brokers prepare submission packs and ensure that data flows smoothly is an obvious opportunity for efficiency gains.

Post-pandemic

An online network that brings together reinsurers, brokers and cedants is another important part of Supercede's platform that has only grown in importance during the pandemic.

While many in the London market continue to emphasise the importance of face-to-face relationships, even after Covid-19, the Supercede duo argue that there is too much serendipity involved in the status quo.

Any client won through a chance encounter at a conference is also a deal that may just as easily never have happened. This is particularly true in Asia, where drumming up business in person is an obvious impediment to connecting risk with capacity that may be available in other regions.

Unlike other placement networks, Supercede is independent from existing industry incumbents and adopts a traditional approach to quoting, authorising and signing, which means that it can handle complex, bespoke deals.

Formerly known as Riskbook, Supercede rebranded after taking on £2 million of funding in 2020 and building out its offering to include the new analytics solution.

Last week it joined Lloyd's Lab, the market's insurtech programme, which will provide a forum for feedback across all the syndicates and help to develop the platform for even the most niche lines of business.

"In the long term, I hope that if we lay the groundwork well enough we will actually see reinsurance made easier to buy — and therefore we'll see more reinsurance bought," says Rose. "So who knows, we might even be able to help grow the industry."

That is a goal everyone can get behind.

Dealsuite

Why corporate buyers are embracing the online revolution

Digital tools are revolutionising middle-market M&A.

Finding large volumes of relevant deals is easier than ever before thanks to a new generation of digital tools. These innovative solutions are making it possible for smaller companies to implement strategies such as programmatic M&A, which were once only available to large players.

The traditional approach to M&A relied on old-fashioned networking and endless “catch-up” meetings to maintain brand awareness and secure a pipeline of potential acquisitions. This approach has always worked much better for large corporations. Smaller corporate buyers either missed out on the best opportunities or had to devote disproportionate resources just to get a foot in the door.

This is changing fast and is even accelerating in the wake of the Covid-19 pandemic. M&A is no longer about who you know — it is about the tools you use. And, more important, those tools are now available to a much wider audience.

Leveraging matching algorithms, cloud services and secure communications, online platforms are introducing new levels of efficiency to the M&A process. That means deals move faster, more smoothly and are less expensive to complete. It also means they fail less often.

Brand pull

Amplifying your M&A brand online provides the best opportunity to bring your ambition, capital and skills to the table. This is one of the main reasons why companies are embracing digital platforms to take their M&A to the next level.

There is an obvious network effect from such platforms that works to the benefit of everyone involved. Connecting with an international community of active dealmakers who are always on the lookout for the next deal maximises the chances of success for corporate M&A teams. It also reduces many of the costs, frictions and pain points that have hampered M&A in the past.

Whether looking to buy or sell an asset, access to a highly engaged network of brokers, investment firms and other corporate M&A teams ensures visibility for your M&A brand.

Companies can also showcase their leadership through such platforms. Dealsuite members can take part in sector meetings that draw on the largest M&A community in Europe, with more than 4,000 engaged M&A professionals.

Market intelligence

Active acquisition strategies are supported by market intelligence that allows M&A teams to spot industry trends, analyse and compare potential targets, and refine opportunities. Armed with clearly defined investment interests and criteria, companies can gain access to a large volume of relevant dealflow.

Teams are notified every time a deal comes to market that matches their criteria. This can include everything from sector and geography to more finely tuned screens that look at earnings, growth and details about the size of stake on offer and type of transaction.

Matching algorithms guarantee that only the most relevant deals are placed in front of corporate buyers. This technology makes it easy to review opportunities and stay abreast of the market with minimal effort.

This is allowing M&A teams to become much more strategic about M&A, even at the SME level. Instead of throwing all their eggs into one basket with large, heroic acquisitions, companies are using these real-time analytic tools to identify smaller opportunities. This means they can acquire assets that are much more targeted, easier to integrate and immediately add value — and they can do so more often.

Agility is key in the modern M&A marketplace and accurate intelligence is essential in providing the confidence to move quickly on opportunities when they arise.

Communication

Of course, finding opportunities is only the first step. Open channels of communication are also vital to converting those potential opportunities into deals. This is where platforms such as Dealsuite.com come into their own, providing proprietary, exclusive contact about potential deals. Corporate M&A teams can enter into direct communication with the right people about relevant deals.

This also works in the opposite direction. Based on a company’s listed M&A preferences, sellers and their advisers can reach out to potential acquirers or investors with proposals directly.

A sophisticated M&A strategy will also involve divestitures and the ability to conduct direct market outreach to relevant professionals is a valuable resource. This can significantly reduce the time it takes to close a sale, minimising the cost and disruption of shedding non-core assets.

Execution
Once a deal is underway, getting it across the line efficiently is the priority. The best-in-class data room from Admincontrol by Visma is the industry's most trusted resource for managing transactions. It is available to Dealsuite members, providing a user-friendly and secure platform for due diligence, document review and secure messaging.

"We recommend Admincontrol's data rooms to our clients because it is a highly secure data room," says Kristin Bratengen, director at Oaklins. "Both the administration and monitoring of the data room is simple and it is also very easy to use for first time users."

Closing
One of the founding principles at Dealsuite was to make the M&A process more effective so that buyers and sellers can close more deals. As an active community founded by M&A professionals and advised by a board of industry specialists, the platform is continuously being developed with security at its core.

At Dealsuite, we are your trusted partner for the future.

Finding FinCrime when we don't know what to look for

Financial criminals are becoming increasingly sophisticated, and so are the tools needed to catch them. Trust and transparency are key to successful deployment. Models that emphasize explainability and auditability can act as virtual teammates, working in partnership with their human operators.

Artificial intelligence is an essential tool in the fight against financial crime, but how do we learn to trust the AI super sleuth?

Understanding is the key. Fundamentally, AI is a crime-fighting partner, not a black box. Knowing how AI models work to detect FinCrime can help to demystify some common misconceptions and build trust.

Domain knowledge

The starting point for any good model is a basic understanding of FinCrime. To ensure quality outputs, we need to make sure that the model is working with good input data. Domain knowledge is critical to this process.

“Today’s approaches to ML and AI are generally domain-agnostic, ignoring domain knowledge that extends far beyond the raw data itself,” said the authors of the Argonne National Laboratory paper AI for Science. “Improving our ability to systematically incorporate diverse forms of domain knowledge can impact every aspect of AI.”

Anomaly detection is at the core of FinCrime models and relies on the incorporation of such contextual knowledge. This starts with the basic characteristics of financial data, such as transaction size, currency, time, and location, through to more detailed characteristics and statistical relationships.

Machine learning helps to identify anomalies far more efficiently than is otherwise possible. Data scientists decide what features to input into a machine learning model, but they do not decide which features are important to the model. Anomaly detection models can determine the important features on their own.

Knowing that the model has been trained on high-quality data that has been prepared appropriately gives operators the confidence they need to trust its outputs.

Proprietary risk factor and pattern repositories

Because we can use the same risk factors and patterns with multiple financial institutions, the repositories we use grow over time.

We develop these factor repositories with domain experts. One example is Benford’s Law, which can be a useful statistical technique to flag questionable financial transactions. To put it simply, Benford’s Law says that the numeral 1 will be the leading digit in a genuine data set of numbers 30.1% of the time, while the numeral 2 will be the leading digit 17.6% of the time. Each subsequent numeral through 9 will be the leading digit with decreasing frequency.

This produces a distinctive distribution — it looks like a slide. These consistent distributions appear in real-world data sets across domains, from stock prices to likes on social media. However, Benford’s Law doesn’t apply universally. It’s important to understand the nature of the data set and the expected distribution of numerals. At Hawk AI, we have observed that many merchant transactions follow Benford’s Law, so we added it as a risk factor. Now the model can detect transactions that don’t follow these distributions and raise alerts.

Positive feedback loops

AI models learn as they are corrected. As more cases are reviewed by operators, the model gets better at detecting FinCrime. In this sense, AI isn’t replacing human operators but working in partnership with them — a virtual teammate.

This is fundamentally different to traditional rules-based screening and transaction monitoring, where human operators repeatedly correct the same mistakes with little or no improvement in the overall result. As well as being fruitless, it can be boring, repetitive work.

With an AI model, the human operators are constantly improving the quality and accuracy of the output. The decisions that operators make are incorporated into the model itself. This doesn’t just produce better results for the model, but also helps the human operators to trust the results of AI and machine learning models. They know that their subject matter expertise played an important role in the results.

Explainability
Financial regulators are increasingly stressing the importance of explainable AI, and rightly so.

Trusting the outputs produced by an AI model requires that users can understand how it makes decisions. Indeed, this is a broad principle of AI in all sectors and not only FinCrime. As AI becomes more complex, rules around transparency and responsible development are a topic of growing significance.

At Hawk AI, our models produce natural language explanations for each risk factor when an anomaly is detected. In its role as a team member, the model supports the investigation by allowing human operators to look into specific transaction and risk factor groupings. The model generates these explanations in plain language that anyone can understand. Operators don't need to be specialists in data science to understand why the AI identified specific activities as suspicious.

Auditability
AI should be auditable as well as explainable, particularly as regulations around the use of AI in finance continue to develop. If a financial institution can't rely on AI solutions to provide results that regulators can audit, the solution isn't much use at all.

Auditability is also a powerful and important component of trust. Not only should a good model explain itself in plain language, but every decision it makes should also be logged and reproducible. This is a core feature of Hawk AI's models. Tracking artifacts show how the model was trained, why it made a certain decision, and why it gave the score it did. If an auditor asks a financial institution these questions, the model will provide the answers.

Model validation
Before any model is deployed, it goes through a validation process, which looks like this:

- Stage 1: Set strict criteria for model deployment.
- Stage 2: Set performance benchmarks for models.
- Stage 3: Compare new model iterations to previous models.
- Stage 4: Make adjustments based on the model's performance.

When FinCrime investigators know how a model was developed and validated, they can rest assured that it will generate reliable results.

Building trust
The effort that goes into developing and validating models creates the foundation for strong and effective partnerships with operators. Together, AI super sleuths and their human teammates can detect anomalies with unmatched speed, accuracy, and transparency.

With criminals now using their own AI, financial institutions of all sizes need to take the threat seriously. Rules-based transaction monitoring systems can't keep pace with the development of modern FinCrime. Catching these criminals requires a sophisticated suite of tools to detect anomalies as they happen. AI models can learn the techniques and spot them in the future, as well as detecting previously unknown risks, all while explaining what they're doing and why. All of this results in AI models that operators can trust to detect financial crime.

Client: Klarna

Open banking makes game-changing ideas powerfully real

Conor Tiernan, UK & Ireland lead for Klarna Kosma, recently shared some of his thoughts on how open banking is changing the way that online businesses operate.

Open banking is making payments cheaper and faster, and enabling businesses to get better insights into customers' banking data to make better decisions.

Online businesses can take payments from customers in many ways. But some of them are quite costly. That's where being able to embed a pay-by-bank solution quickly and easily into your website can cut costs and get the funds to you sooner, meaning you can deliver goods or services quicker.

How businesses benefit from open banking

Let's say you're an insurance provider. To provide the best policies to your customers, you want to understand more about them. Letting them quickly and easily connect their bank account is a great way to do this. You can understand their spending history and get a clearer picture of their financial habits, which you can use to arrive at a quicker and better-informed decision. And that's good for customers too. As an insurer, you want to make sure that your customers always have the right amount of cover for their needs. With open banking, you can make this happen.

If it sounds too good to be true…

Of course, there are some challenges when it comes to implementing open banking. These are the big ones:

- **Going global.** When you start in your home market, it's relatively straightforward to connect to the banks. But it's a completely different ballgame as soon as you step foot in the international market. You need to consider all the different countries and banks you need to connect with and there are additional licenses that you need to look at as well. It can be a huge headache.
- **Integration.** Integration is not necessarily too complex on a bank-by-bank basis, but there are potentially 6,000 banks that you need to connect with across the EU alone. That's where the integration side can get complex quickly.
- **Raw data.** Once you've built out your international strategy and integrated with all the banks you want to, now what? You've got a bunch of raw banking data, but that's not very useful by itself. What you want is enriched banking data, so now you've got another major project to undertake before you end up with something useful.

Solving the challenges

Working with a partner that's got a long track record is the key to overcoming all these challenges.

Klarna Kosma is unique because we've been connecting to banks since 2005 — and connecting to banks is what underpins open banking. We've got a huge amount of experience. We process hundreds of millions of bank payments each year over open banking. We enable customers to connect their bank accounts and carry out billions of open banking data sessions.

When you start working with Kosma, what do you get from day one? We give you the gateway to the world. We enable you to connect with bank accounts globally, so that you don't have to. Kosma has taken care of all these headaches for you.

Furthermore, our single API gives you a quick and seamless integration, meaning that you can reduce your costs because you don't need an army of engineers to maintain all these costly bank connections.

On top of that, we've built this enrichment layer that gives you the ability to provide insights. We've got an intelligence engine that gives you the categorization, the insights, and helps you generate reports.

All this is something that we've understood ourselves after being around so long. And that's why we put it into play and that's why Kosma is the go-to partner for open banking.

What is Klarna Kosma?

Kosma itself is an open banking provider. You might be interested to learn that we also provide the infrastructure to other open banking providers. Why would we do that? We do it because we have the widest reach in terms of bank connections and we want to enable others to bring wonderful open banking solutions to market. We provide the infrastructure, we've got the experience, we've got the products, and we've got a single API that opens the door for you as a business, a lender, or another open banking provider.

Improving the customer journey with embedded finance

Getting the customer experience right is critically important to building a scalable B2B e-commerce platform.

You can have the right product, at the right price, delivered to the right audience, but it might all count for nothing if you don't make the customer journey as painless as possible.

Zero friction
This has long been understood in the consumer world. Companies like Amazon have invested millions of dollars in getting rid of every possible point of friction that stands in the way of a sale. Their research into why customers abandon shopping carts has revealed a lot about how to build a zero-friction customer journey — and many of those lessons are just as valid in the B2C world.

Whether customers are shopping for sunglasses or freight forwarding, they want to be able to find the right product and pay for it with as little hassle as possible. And that last part has been one of the biggest obstacles to B2B e-commerce in the past. Building a website is easy, but offering the kinds of flexible payment terms that business customers are used to is much more difficult.

Flexible payments
Buyers who typically settle invoices after 30 days want the same flexibility when ordering online. A common solution has been to take a hybrid approach, allowing buyers to order online but asking them to call the merchant or submit documents if they want flexible payment terms.

You don't need to be Jeff Bezos to see how this creates friction. Every extra step the customer has to take will result in missed revenue. And vice versa. Every step you can get rid of will improve your sales conversion numbers.

Just like buy now-pay later services on B2C websites, the solution for B2B platforms is to embed net payment options at the point of purchase, making it as convenient as possible for the buyer to complete a transaction using data they have already provided through their onboarding and purchasing history. Instead of having to go to an insurance broker to insure the transaction or a factoring company to get an advance, everything happens on the merchant's website.

This has several benefits:

- **Less friction** – Removing the number of manual steps in the buying process makes buyers more likely to convert and to keep coming back.
- **Improved loyalty** – Providing access to a line of credit helps to build a trust relationship between buyer and seller, which improves customer loyalty to your business. We see buyer retention grow by 40%.
- **More customers** – Combining a variety of data sources improves the ability to screen buyers and increases the acceptance rate for payment terms.

At Sprinque we offer all the above functionality through a simple integration. In addition, by paying the marketplace or merchant at the time of the final invoice, we reduce the risk of default and free up valuable working capital.

Client: Fourthline

Fourthline

Finding the sweet spot

How to strike the right balance between compliance and conversions.

INTRODUCTION

Striking the right balance between customer experience and a fully compliant onboarding process is an increasingly tough challenge for fintechs. Organizations focused on agile growth often seek to prioritize a seamless user journey that maximizes conversions while minimizing friction. Getting it right is difficult.

Onboarding customers while satisfying requirements for anti-money laundering (AML), fraud screening, data protection, and account authentication can be complex, expensive, and negatively impact the user experience. Inevitably, compliant processes add friction and detract from fintechs' ability to respond and adapt to market opportunities.

Customers care about security and privacy but also have high expectations regarding user experience. Is it possible to have it all?

At Fourthline, we have demonstrated experience that it is. In this Whitepaper, we outline the common challenges fintechs face when balancing compliance and conversions and our recommendations for finding the sweet spot.

Common challenges

There are many ways for fintechs to strengthen their compliance, but most methods are complex and lead to a variety of negative outcomes, such as:

- Slow onboarding leading to customer dissatisfaction
- Poor conversion rates leading to a lack of competitiveness
- Colossal compliance investments leading to business rigidity
- Strategic inflexibility slowing growth and scaling goals
- Risk exposure leading to fines and reputational damage

In this section, we've laid out **today's three most significant challenges** when balancing compliance and conversion.

CHALLENGE 1: Extreme regulatory complexity leading to fines and reputational damage

Identity verification is not a single event anymore; it's a journey from customer onboarding and authentication to account recovery. Know Your Customer (KYC) onboarding helps to prevent financial crimes and protect against fraud. Anti-Money Laundering (AML) and fraud screening are designed to detect and prevent financial crimes. Account authentication helps to ensure that only authorized individuals have access to customer accounts. Finally, re-KYC helps fintechs periodically verify their customers' identities to ensure that their information is up-to-date.

On top of KYC-AML-focused regulations, European fintechs must comply with EU regulations such as the Second Payment Services Directive (PSD2) and the General Data Protection Regulation (GDPR), among others.

For fintechs, balancing conversions and meeting all these regulatory requirements is a considerable challenge, as it is time-consuming, costly, and often requires extra resources and investment.

Additionally, in Europe, each country has defined specific rules. This situation creates immense regulatory fragmentation. It adds complexity for fintechs seeking to comply with the rules of several member states and usually ends up slowing down their European expansion

Lastly, governments and regulatory authorities are constantly updating and revising these regulations. Fintechs need to stay updated with these changes to avoid huge fines or reputational damage. However, as relatively new players in the financial services industry, they often don't have the same resources or knowledge to keep up with these changes.

CHALLENGE 2: Colossal compliance investments leading to strategic rigidity

Implementing a streamlined and efficient KYC process is one of the most critical steps in balancing conversions and compliance. This can be achieved using technology such as Ai-powered ID verification, which has proven to speed up the account opening process and drive conversions while ensuring compliance with regulations.

Implementing such technology requires a significant investment in IT, compliance, and legal expertise. This is a challenge for fintechs, particularly the ones with ambitious time-to-market and operational efficiency goals.

Moreover, balancing conversions and compliance in this area can also be challenging as **new technologies and standards constantly evolve and are costly to implement**. For example, a neobank may have to implement the latest security standards to comply with EU regulations and to secure customer data, which will lead to added costs, time, and effort.

Consequently, **such major investment impacts fintechs' flexibility.** Implementing a streamlined and efficient KYC process does require an organization to attribute a lot of resources to fraud prevention. Unless its financial and people resources are vast, this move can impact the organization's ability to rapidly change its strategy or adapt its risk appetite when necessary.

CHALLENGE 3: Poor conversion rates leading to a lack of competitiveness

High conversion and high growth ambitions trigger a lot of fraud. This is why European fintechs must balance the need to make their digital platforms user-friendly and easy to navigate with the need to ensure that the platforms are secure.

The challenge? Adding too many security measures can make the platform less user-friendly and deter potential customers. The more steps in a conversion funnel, the more opportunities customers have to drop off.

Fintechs must also set up a secure environment to protect customer data and funds. It's not just about identity fraud at onboarding. Making sure their customers' data and funds are protected can also add too many security measures, making the platform less user-friendly. At the same time, too few security measures can result in hefty fines and a loss of customer trust.

The most common misconception about introducing identity verification is that it increases hassle and reduces conversion. While too much friction is bad for your conversion rate, a well-designed user experience that is simple for customers to complete will not cause visible enough difficulty to discourage new customers. So, rather than being a must, if done correctly, KYC verification and account authentication may be used to deliver better service to your customers.

COMPLIANCE REQUIREMENTS

The continuing battle against financial crime is the main driver of regulation. Criminals constantly imagine new innovative methods of fraud and money laundering. Consequently, EU member states have to adapt their regulatory framework. European financial institutions have been challenged to strengthen customer identification processes and customer relationships across the whole lifecycle.

Throughout Europe, regulators require financial institutions to adopt robust procedures to identify customers, assess potential risks, and perform continuing risk monitoring.

This creates a compliance and operational burden that requires fintechs to put in place:

Proper identity verification
Financial institutions are under growing pressure to verify identities and ensure that customers are who they say they are.

Effective watchlist screening
Failure to check customers properly against lists of politically exposed persons, sanctions, and adverse media reports carries potentially significant penalties.

Strong fraud detection
Financial criminals are constantly developing new techniques, making it increasingly difficult to protect against fraud.

Comprehensive customer due diligence monitoring
The requirements above don't stop with onboarding. Financial institutions need to continue to monitor customers over time.

European governments and regulatory bodies have taken different approaches to how they want financial institutions to implement these compliant requirements. The following section focuses on Germany, France, and the United Kingdom to highlight the differences in approach from one country to another.

COMPLIANCE CHECKLIST

GERMANY

AML: Anti-Money Laundering Directive (GwG), EU AML Directives

Data protection: Federal Data Protection Act (BDSG), EU GDPR

Electronic payments: Payment Services Supervision Act (ZAG), EU PSD2

While AML rules in most of Europe are broadly based on the same EU directive, there are differences in how they are implemented. The German regulator in particular is more conservative than its counterparts in other European jurisdictions. As a result, video verification is the most commonly used KYC method.

Identification methods that meet national requirements in Germany include:

- In-person identification (Post Ident)
- Video verification (Video Ident)
- Electronic proof of identity (eID)
- Qualified electronic signature (validated through an electronic payment)

General KYC requirements include:

- Identify the customer
- Evaluate and monitor the nature of the business relationship
- Assess the customer's risk including establishing if the customer is a politically exposed person

FRANCE

AML: Monetary and Financial Code, EU AML Directives

Data protection: Act No 78-17, EU GDPR

Electronic payments: Law No 2000-230, EU PSD2

The French regulatory system prescribes a relatively limited list of approved forms of identitification that can be used for digital onboarding. While this approach is more flexible than the German system, it is still complex and less straightforward than the UK's regime.

According to the National Cybersecurity Agency of France (ANSSI), digital identity can be verified directly using:

- Electronic proof of identity (FranceConnect+, l'Identité Numérique)
- Providers certified by the ANSSI
- Software certified by ANSSI

Otherwise, identity can be verified by combining at least two of the following measures:

- Obtain a copy of an official identity document
- Verify and certify an official document through an independent third party
- Require first payment be made with an account opened in the client's name
- Confirm the customer's identity through an approved third party
- Use an ANSSI-compliant service to verify the identity
- Collect a qualified electronic signature

UNITED KINGDOM

AML: Proceeds of Crime Act 2002; The Money Laundering, Terrorist Financing and Transfer of Funds (Information on the Payer) Regulations 2017

Data protection: Data Protection Act, UK GDPR

Electronic payments: Payment Service Regulations

The Joint Money Laundering Steering Group indicates in their AML guidance that additional verification checks should be applied to manage the risk of impersonation fraud in case of non-face-to-face KYC, such as:

- Verify biometric data
- QES
- One cent transaction (UK or EU)
- Telephone contact
- Mail by post
- Internet sign on using security codes/tokens provided by mail to a verified address
- Certify copy of ID document

The UK regulator takes a risk-based approach to KYC, requiring that firms have sound processes in place, but there are no laws or rules that prescribe specific measures.

Broadly speaking, fintechs need to meet three basic KYC requirements:

- Identify the customer
- Verify the customer's identity
- Assess and monitor the nature of the business relationship

BEST PRACTICES

Few fintechs prioritize compliance early in their journey when the focus is usually on product development and growth. Even when they get around to thinking about it, too many see compliance as a box-ticking exercise and implement low-quality solutions that don't perform well.

Unfortunately, this approach leads to hefty fines, reputational damage, costly legal battles, and business closure.

To avoid such pitfalls, fintechs can implement the following three best practices:

Develop a compliance mindset early
While regulatory scrutiny may be lower in the early stages of a fintech's growth, good compliance practices will yield dividends as the business grows. It also helps you get ahead of the curve when compliance requirements intensify.

Focus on efficiency and flexibility
Implement solutions that can scale quickly in line with growth plans, have low up-front costs, and don't incur high expenses for unused capacity.

Harness automation, technology and outsourcing
Don't reinvent the wheel. Take advantage of solutions that address your operational needs, allowing you to deploy your resources where they can add the most value.

Fintechs that follow these principles have proved that an onboarding process can be efficient, seamless, and simple while meeting all regulatory requirements and preventing fraud.

What's the best way for fintechs to implement these best practices? Working with the right partner.

CHOOSING A PARTNER

The key to maximizing conversions is working with a provider who understands your needs. So what should you consider as a European fintech if you're looking for a compliance partner?

United
The specialist you choose should work with your teams as a true partner rather than a KYC provider. It should help you meet all your compliance needs at onboarding and across your clients' entire lifecycle. It should take care of your compliance so you can focus on your business.

Secure
Your partner should guarantee excellence in fraud prevention without compromising on conversion.

Scalable
You don't need to trade growth for compliance. Your partner should offer a dynamic solution that adapts to your specific needs as you grow in size and/or expand into new markets.

Ready-to-use
Partnering with a specialist that offers a single API that integrate seamlessly with your technology will ensure your business complies with all regulations without navigating multiple systems or providers. It also helps to reduce the time and cost associated with compliance.

Streamlined
Partnering with a specialist that offers a service designed with the customer in mind will ensure your organization provides the smoothest customer experience and better conversion rates.

At Fourthline, we have demonstrated experience that prioritizing these points helps European fintechs increase their growth and fraud prevention while strengthening resilience and flexibility.

Get in touch with our experts to learn more about our approach and what our partnership could do for you and your business.

Client: Sygno

Sygno

Modelling good behaviour

Focusing on good customers is key to a more effective transaction monitoring process.

Transaction monitoring has become an endless arms race that's costly, cumbersome and produces very little in the way of results.

It's time for a rethink. Instead of fighting a losing battle trying to model criminal behaviour, as we're used to, why don't we model the behaviour of good customers?

Training a model to catch financial criminals is like trying to find a needle in a haystack — except the needle is constantly changing to avoid being found. A much more effective approach is to train your model to recognise the hay.

After all, there's substantially more hay than needles and it has no incentive to change, making it much more predictable. The model therefore doesn't need to know what needles look like — it just needs to know whether something isn't hay.

It's easy to be surprised by new and emerging types of crime when you're only focusing on known criminal phenomena. Criminals are trying not to get caught!

This is why we talk about modelling good behaviour. What can you expect in a situation from a totally legitimate customer?

Look at the haystack, not the needle

Instead of trying to spot criminals, this approach focuses on normal customer behaviour with a model that's trained on legitimate transaction data.

This might sound simple, but it works. Looking at customers instead of criminals shines a light on previously undetected patterns. Banks that have implemented this type of model have often substantially increased their number of previously undetected crimes and cut their false positives by 80%.

This rethink also brings other benefits. Instead of relying on dozens of incomprehensible rules, being tweaked and tuned over time, this risk-based approach is more straight forward and therefore easier to explain to regulators. Think white box, not black box.

Compliance costs

The stakes are high. Most news on transaction monitoring has been about banks incurring fines, but that's only the tip of the iceberg. LexisNexis estimates the global cost of AML and sanctions compliance to be more than $210bn a year. Most of this money is spent on cumbersome administrative processes targeting totally legitimate customers while doing their daily business.

Despite these staggering efforts, criminals continue to thrive by finding the loopholes in the system. It can seem impossible to keep pace with how they're operating, leading to an endless cat-and-mouse game with very limited (and expensive) results.

By focusing your model on the good customers, you can once again re-focus your efforts on the real criminals, rather than the other way around.

Start modelling good behaviour today

Want to know more about how this approach can increase protection and drastically reduce false positives of your existing monitoring system? Get in touch and we'll be happy to discuss solutions.